AWS Infrastructure Cost Estimation

Development Environment

Cargo Connect

AIT (AI Transformation) Platform

Gold Star Foods Group

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Executive Summary

This document provides a comprehensive AWS infrastructure cost estimation for the AIT (AI Transformation) Development environment at Gold Star Foods. The estimate is based on detailed workload assumptions and AWS Pricing Calculator outputs.

Cost Overview

Cost Category	Amount (USD)
Upfront Cost	$30.00
Monthly Cost	$1,477.75
Annual Cost (12 months)	$17,763.00

Top Cost Drivers

Service	Monthly Cost	% of Total
AWS Fargate (ECS)	$865.06	58.5%
AWS DMS	$240.39	16.3%
Amazon VPC	$159.51	10.8%
Amazon Route 53	$100.52	6.8%
Aurora PostgreSQL	$64.40	4.4%
Other Services	$47.87	3.2%

Development vs Production Comparison

Configuration	DEV	PROD	Savings
Monthly Cost	$1,477.75	$4,302.57	$2,824.82 (66%)
Fargate Tasks	1 task, 2 GB	2 tasks, 4 GB	$2,594.60
Aurora ACUs	1 ACU	4 ACUs	$161.70
NAT Gateways	1 (Single AZ)	2 (Multi-AZ)	$66.89
CloudFront Transfer	2.7 GB/month	18 GB/month	$1.63

Infrastructure Assumptions

The following assumptions were used as inputs for the AWS Pricing Calculator to generate the cost estimates for the Development environment.

General Parameters

Parameter	Value
Database Storage	~150 GB
Monthly Work Hours	160 hours
React SPA Size	~100 MB
Dev Monthly Deployments	~30 (more frequent than prod)
CloudFront Cache Misses	~135 MB (15% of prod)

Database Configuration (DEV)

Metric	Value
Aurora Capacity Units (ACUs)	1 (vs 4 in PROD)
Storage Amount	150 GB
Backup Storage	150 GB
Pricing Strategy	On-Demand

Compute Configuration (DEV)

Metric	Value
Fargate Tasks	1 per day (vs 2 in PROD)
Memory Allocated	2 GB (vs 4 GB in PROD)
Ephemeral Storage	20 GB
Average Duration	24 days

Network Configuration (DEV)

Metric	Value
NAT Gateways	1 (Single AZ)
Availability Zones	1 (vs 2 in PROD)
Public IPv4 Addresses	1
Data Transfer (Intra-Region)	4,500 GB/month

Detailed Cost Breakdown

The following table provides a complete breakdown of all AWS services included in this estimate.

Service	Upfront	Monthly	Annual
AWS Fargate (ECS)	$0.00	$865.06	$10,380.72
AWS Database Migration Service	$0.00	$240.39	$2,884.68
Amazon VPC (NAT Gateway, IPv4)	$0.00	$159.51	$1,914.12
Amazon Route 53	$0.00	$100.52	$1,206.24
Aurora PostgreSQL Serverless	$0.00	$64.40	$772.80
Amazon CloudWatch	$0.00	$45.92	$551.04
AWS Secrets Manager	$0.00	$1.20	$14.40
Amazon ECR	$0.00	$0.40	$4.80
Amazon CloudFront	$0.00	$0.28	$3.36
AWS Systems Manager	$0.00	$0.05	$0.60
Amazon S3	$0.00	$0.02	$0.24
Amazon Cognito	$0.00	$0.00	$0.00
AWS Certificate Manager	$30.00	$0.00	$30.00
TOTAL	**$30.00**	**$1,477.75**	**$17,763.00**

Important Notes

⚠ **Disclaimer**

This document provides preliminary cost estimates based on projected usage patterns and is subject to change. Actual costs may vary depending on real-world usage, AWS pricing updates, and infrastructure adjustments.

Note: This estimate does not include Amazon Bedrock usage costs, which are provided in a separate document.

- All prices are in USD and based on US East (N. Virginia) region pricing.
- DEV environment uses reduced resources compared to PROD for cost optimization.
- Single AZ deployment (no Multi-AZ redundancy) - acceptable for development workloads.
- DMS configuration remains Multi-AZ to ensure data integrity during migrations.
- Route 53 cost is shared between Development and Production environments.
- CloudWatch and monitoring configuration mirrors PROD for consistent observability.
- Cognito costs assume up to 1,000 monthly active users (MAU) within the free tier.